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LIMIT THEOREMS FOR THE NUMBER OF OCCURRENCES OF CONSECUTIVE k SUCCESSES IN n MARKOVIAN TRIALS

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Abstract

We present a method of deriving the limiting distributions of the number of occurrences of success (S) runs of length k for all types of runs under the Markovian structure with stationary transition probabilities. In particular, we consider the following four best-known types. 1. A string of S of exact length k preceded and followed by an F, except the first run which may not be preceded by an F, or the last run which may not be followed by an F. 2. A string of S of length k or more. 3. A string of S of exact length k, where recounting starts immediately after a run occurs. 4. A string of S of exact length k, allowing overlapping runs. It is shown that the limits are convolutions of two or more distributions with one of them being either Poisson or compound Poisson, depending on the type of runs in question. The completely stationary Markov case and the i.i.d. case are also treated.

RUNS; MARKOV CHAINS; STATIONARY TRANSITION PROBABILITIES; CONVERGENCE; POISSON; COMPOUND POISSON; RELIABILITY; CONSECUTIVE-k-OUT-OF $n:F$ SYSTEM

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1. Introduction

Let $n > k \geqq 1$ be two fixed positive integers. For a given sequence of n randomly arranged S (success) and F (failure), there are many ways of counting the number of *runs of S of length k*. Four of the best-known types are:

I. A run of S of length k means a string of S of exact length k preceded and followed by an F, except the first run which may not be preceded by an F or the last run which may not be followed by an F.

II. A run of S of length k means a string of S of length k or more.

III. A run of S of length k means a string of S of exact length k, where recounting starts immediately after a run occurs.

IV. A run of S of length k means a string of S of exact length k, allowing overlapping runs.

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Let N_{I} be the number of occurrences of consecutive k successes of Type I. The other three variables N_{II}, N_{III} and N_{IV} are defined analogously for the other three types. (For brevity we suppress the dependence of these variables on k and n.) For example, consider the following realization, with $n=16$:

$$\text{SSSSSSFFFSSSFFSS}.$$

If we take $k=3$, then $N_{\mathrm{I}}=1$; $N_{\mathrm{II}}=2$; $N_{\mathrm{III}}=3$; and $N_{\mathrm{IV}}=5$. In the recent paper of Wang and Liang (1993), an algorithm for computing the probability distributions under the Markovian structure with stationary transition probabilities of the four variables is proposed. For a brief discussion of the four types of runs, such as their applications and their relations with other problems, we refer the reader to that paper.

The purpose of this paper is to give a unified approach to the derivation of the limiting distributions of all the four counting variables for *all* $k \geqq 1$ as n tends to infinity and some minor conditions. We shall show that under the Markovian structure with stationary transition probabilities the limits are convolutions of two or more distributions with one of them Poisson for N_{I} and N_{II} and compound Poisson for N_{III} and N_{IV}. Similar phenomena occur under the completely stationary Markovian structure. But under the independent and identically distributed (i.i.d.) structure, all the limits are Poisson.

Throughout this paper, we adopt the usual convention of denoting S by 1 and F by 0. Let $X_1, X_2, \cdots, X_n$ be a sequence of Markov Bernoulli random variables with stationary transition probabilities

$$(1) \qquad \begin{array}{cc} & \begin{array}{cc} 0 & 1 \end{array} \\ \begin{array}{c} 0 \\ 1 \end{array} & \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix} \end{array} \qquad (0<\alpha, \beta<1)$$

and the initial probability

$$P(X_1=1)=p=1-P(X_1=0) \qquad (0 \leqq p \leqq 1).$$

This model contains two special cases.

1. If $p=\alpha/(\alpha+\beta)$, then $\{X_i\}$ is completely stationary in the sense that $P(X_i=1)=p$ for all $i=1, 2, \cdots$, and with transition probabilities

$$(2) \qquad \begin{array}{cc} & \begin{array}{cc} 0 & 1 \end{array} \\ \begin{array}{c} 0 \\ 1 \end{array} & \begin{pmatrix} 1-(1-\pi)p & (1-\pi)p \\ (1-\pi)(1-p) & (1-\pi)p+\pi \end{pmatrix} \end{array},$$

where π is the correlation coefficient of X_i and X_{i+1}. In this model, if $\pi \to 0$ as $n \to \infty$, we call it the *asymptotically i.i.d. Markov Bernoulli model.*

2. If $\alpha=p$ and $\beta=1-p$, then $\{X_i\}$ is the i.i.d. Bernoulli sequence. (The parameters α, β, p and π depend on the index n. For brevity, we shall suppress this index throughout this paper.)

Historically, Koopman (1950) was the first person to take on the problem of finding limiting distributions of the number of runs under a Markovian structure. He obtained the limiting distribution of N_{III} with $k=1$ for the asymptotically i.i.d. Markov Bernoulli model. Later, Dobrušin (1953) obtained many interesting limit results for the same model. Today this model is still being pursued by many authors. See Godbole (1990) and the references cited there. Pedler (1978) was possibly the first person to consider the non-asymptotically independent Markov Bernoulli model (1) in his studies of the limiting distribtions of N_{III} (and others) also for $k=1$. The papers by Wang (1981), Gani (1982), Bühler (1989), and Wang and Bühler (1991) were concerned with the limiting distribution of N_{III}, also for $k=1$, of the completely stationary model (2).

For $k \geqq 2$, some results on the limiting distributions of N_{II}, N_{III}, and N_{IV} in the i.i.d. case can be found in Papastavridis (1987), Goldstein (1990), Godbole (1990), Barbour et al. (1992) and Fu (1993). The recent article by Koutras and Papastavridis (1993) deals with related problems in the i.i.d. case.

To derive our results, we shall take an approach different from all the approaches used by everybody else on this topic. Our idea can be traced back to Doeblin (1938). Instead of directly working with the original Markov Bernoulli sequence, we construct an equivalent parallel sequence for which the limiting distributions of all the four types of runs are obtained. The problem is thus to make sure that both sequences are asymptotically equivalent.

2. The preliminaries

Define two sequences of random variables $\{U_i\}$ and $\{V_i\}$ by

$$U_i = \text{the } \textit{length} \text{ of the } i\text{th sojourn of } \{X_i\} \text{ in state } 1,$$

$$V_i = \text{the } \textit{length} \text{ of the } i\text{th sojourn of } \{X_i\} \text{ in state } 0.$$

The Markovian property and the stationarity of the transition probabilities of $\{X_i\}$ ensure that *all* the components of $\{U_i\}$ and $\{V_i\}$ are mutually independent with marginal geometric distributions:

$$\text{(3)} \qquad \begin{cases} P(U_i=k)=(1-\beta)^{k-1}\beta, & k=1,2,\cdots \\ P(V_i=k)=(1-\alpha)^{k-1}\alpha, & k=1,2,\cdots. \end{cases}$$

Evidently observing the sequence $\{X_i\}$ is the same as observing the alternating geometric process $U_1, V_1, U_2, V_2, \cdots$ if $X_1=1$ and $V_1, U_1, V_2, U_2, \cdots$ if $X_1=0$. Therefore to study the number of occurrences of success runs of the sequence $\{X_i\}$ it is sufficient to study this alternating process. As in Wang (1992), we shall modify the alternating process as follows:

Let $\{\mathbb{U}_i\}$ and $\{\mathbb{V}_i\}$ be two independent sequences of random variables with marginal distributions (3) and $\{\mathbb{U}_i\}$ and $\{\mathbb{V}_i\}$ are also independent of each other. If $X_1=0$, we let $\mathbb{V}_1 = V_1$. When $\mathbb{V}_1$ terminates, we start two runs $\mathbb{U}_1$ and $\mathbb{V}_2$ immediately and simultaneously. We let $\mathbb{U}_1$ run its course, but when $\mathbb{V}_2$ terminates we start another two runs

$\mathbb{U}_2$ and $\mathbb{V}_3$. The process is repeated indefinitely, such that $\mathbb{U}_i$ runs its course, but when $\mathbb{V}_i$ terminates, two runs $\mathbb{U}_i$ and $\mathbb{V}_{i+1}$ are started immediately for all $i \geqq 1$.

If $X_1=1$, let $\mathbb{U}_1=U_1$, and initiate another run $\mathbb{V}_1$ at the same time $i=1$. As before, we let $\mathbb{U}_1$ run its course, but when $\mathbb{V}_1$ terminates, two runs $\mathbb{U}_2$ and $\mathbb{V}_2$ are activated immediately and simultaneously, and the process is repeated indefinitely as described above.

Define

$$W_i^{(\mathrm{I})}=I(\mathbb{U}_i=k), \qquad (I(A) \text{ denotes the indicator function of } A)$$

$$W_i^{(\mathrm{II})}=I(\mathbb{U}_i \geqq k)$$

$$W_i^{(\mathrm{III})}=[\mathbb{U}_i/k], \qquad ([x] \text{ denotes the integral part of } x)$$

$$W_i^{(\mathrm{IV})}=\begin{cases}\mathbb{U}_i-k+1, & \text{if } \mathbb{U}_i \geqq k,\\ 0, & \text{otherwise,}\end{cases}$$

$i=1, 2, \cdots$.

Since $\{V_i\}$, having geometric distribution (3), is *memoryless*, i.e.

$$P(V_i=j \mid V_i>l)=P(V_i=j-l)$$

for all $i=1, 2, \cdots$, and $j>l \geqq 1$, the original alternating process in which V and U alternate is *asymptotically equivalent* to the modified process, provided that $P(V_i>U_i$ for *all* $i \mid i \leqq n) \to 1$ and $n \to \infty$. The latter is in fact the case, if $\alpha \to 0$, as $n \to \infty$. Its proof is simple and can be found in the proof of the next lemma and will not be repeated.

For fixed $n \geqq 2$, define a stopping time by

$$M_n=\begin{cases}0, & \text{if } \mathbb{V}_1 \geqq n,\\ \max\{k: \mathbb{V}_1+\mathbb{V}_2+\cdots+\mathbb{V}_k \leqq n-1\}, & \text{otherwise.}\end{cases} \tag{4}$$

Then M_n is binomial with parameters $(n-1, \alpha)$ and independent of $W_j^{(i)}$ for all $i=$I, II, III, and IV and $j \geqq 1$. Denote

$$\mathbb{S}_n^{(i)}=W_1^{(i)}+\cdots+W_{M_n}^{(i)}. \tag{5}$$

Because in the modified process $\mathbb{U}_i$ may overlap, we have

$$P(\mathbb{S}_n^{(i)} \geqq N_i)=1 \quad \text{and} \quad P(\mathbb{S}_n^{(i)}>N_i)>0,$$

for all finite $n \geqq 2$.

Lemma 1. For all $i=$I, II, III and IV, we have

$$\lim_{n \to \infty} P(\mathbb{S}_n^{(i)}>N_i)=0,$$

if $n\alpha \to \lambda \geqq 0$ as $n \to \infty$. (Note that in model (1) we asumed $0<\alpha$, $\beta<1$.)

Proof. It is easy to verify that for all i

$$P(W_1^{(i)}>\mathbb{V}_1) \leqq P(\mathbb{U}_1>\mathbb{V}_1) \leqq \alpha/[1-(1-\alpha)\beta],$$

so that

$$\begin{aligned}
P(\mathbb{S}_n^{(i)} > N_i) &\leqq \sum_{j=1}^{n-1} P(W_h^{(i)} > \mathbb{V}_h \text{ for some } h=1,\cdots,j \mid N_n=j)P(N_n=j) \\
&\leqq \sum_{j=1}^{n-1} P(\mathbb{U}_1 > \mathbb{V}_1) jP(N_n=j) \\
&\leqq P(\mathbb{U}_1 > \mathbb{V}_1) n\alpha \\
&\leqq n\alpha^2/[1-(1-\alpha)\beta].
\end{aligned}$$

The last term tends to 0 if $n\alpha \to \lambda \geqq 0$ as $n \to \infty$.

3. The results

By Lemma 1, to find the limiting distributions of N_i it is sufficient to find those of $\mathbb{S}_n^{(i)}$.

Denote the probability mass functions (p.m.f.) and probability generating functions (p.g.f.) of $W_j^{(i)}$ to be f_i and g_i, respectively, for $i=$ I, II, III and IV, and all $j=1, 2, \cdots$. It follows from the definitions of the W's that:

(6) $$\begin{cases} f_{\mathrm{I}}(x) = \theta_1^x(1-\theta_1)^{1-x}, & \text{for } x=0, 1, \\ g_{\mathrm{I}}(t) = 1+\theta_1(t-1), & \text{for all } t \in \mathbb{R}, \end{cases}$$

where $\theta_1 = (1-\beta)^{k-1}\beta$.

(7) $$\begin{cases} f_{\mathrm{II}}(x) = \theta_2^x(1-\theta_2)^{1-x}, & \text{for } x=0, 1, \\ g_{\mathrm{II}}(t) = 1+\theta_2(t-1), & \text{for all } t \in \mathbb{R}, \end{cases}$$

where $\theta_2 = (1-\beta)^{k-1}$.

(8) $$\begin{cases} f_{\mathrm{III}}(x) = \begin{cases} 1-(1-\beta)^{k-1} \\ (1-\beta)^{k-1}[(1-\beta)^{k(x-1)}(1-(1-\beta)^k)] \end{cases} & \begin{matrix} \text{if } x=0, \\ \text{if } x=1, 2, \cdots \end{matrix} \\ g_{\mathrm{III}}(t) = 1-(1-\beta)^{k-1} + \dfrac{(1-\beta)^{k-1}-(1-\beta)^k)s}{1-s(1-\beta)^k} & \text{for } s<(1-\beta)^{-1}. \end{cases}$$

And

(9) $$\begin{cases} f_{\mathrm{IV}}(x) = \begin{cases} 1-(1-\beta)^{k-1} \\ (1-\beta)^{k-1}[\beta(1-\beta)^{x-1}] \end{cases} & \begin{matrix} \text{if } x=0, \\ \text{if } x=1, 2, \cdots \end{matrix} \\ g_{\mathrm{IV}}(t) = 1-(1-\beta)^{k-1} + \dfrac{(1-\beta)^{k-1}\beta s}{1-s(1-\beta)} & \text{for } s<(1-\beta)^{-1}. \end{cases}$$

Let G_i be the p.g.f. of $\mathbb{S}_n^{(i)}$. Then by (4), (5), and conditioning on X_1 we have

(10) $$G_i(t) = [pg_i(t)+(1-p)][1+\alpha(g_i(t)-1)]^{n-1}.$$

It is interesting to note that both p.m.f. in (8) and (9) are mixtures of two p.m.f. with one of them Bernoulli. In Wang (1989), Theorem 6, it was proved that if a sequence of i.i.d. random variables $\{X_i\}$ whose common distribution is a mixture of two p.m.f. with one of them Bernoulli, then its partial sum $S_n = X_1 + \cdots + X_n$ has a compound Poisson limiting distribution.

If the limits of the distributions of N_i are to exist, a dominating condition is that the number of transitions from state 0 to state 1 *must* be very moderate, relative to n (as $n \to \infty$). That is to say, $n\alpha$, the mean number of transitions from state 0 to state 1, must approach a constant as $n \to \infty$. We discovered that this is the *only* condition needed to ensure the existence of the limiting distributions of N_i for all k ($\geqq 1$) in the Markovian model (1) and in the completely stationary model (2). (For the i.i.d. model, the condition can be weakened somewhat.) The other parameters p and β in the Markovian model and π, the correlation coefficient of X_i and X_{i+1}, in the completely stationary model play no roles here. They can stay constant or have their own limits, such as $p \to p_0$, $\beta \to \beta_0$, and $\pi \to \pi_0$, for p_0, β_0 and $\pi_0 \in [0, 1]$. (If $\pi_0 = 0$, the completely stationary model (2) is reduced to the asymptotically i.i.d. Markov Bernoulli model.) Therefore, without loss of generality, we shall assume that the parameters p, β, and π stay constant in the next four theorems and Corollary 1. (The condition '$n\alpha \to \lambda$ as $n \to \infty$' and its equivalent '$\Sigma_{i=1}^{n} P(X_{i+1} = 1 \mid X_i = 0) \to \lambda$ as $n \to \infty$' in the non-stationary case was first used by Koopman (1950) and Dobrušin (1953) and has since become a key condition in Poisson approximation problems for Markovian sequences.)

The next four theorems follow from the five equations (6) to (10). They can be proved by replacing each of the four p.g.f. g_i in G_i and letting $n \to \infty$ to obtain the limits ϕ_i, for $i = \mathrm{I}, \mathrm{II}, \mathrm{III}$, and IV. The demonstrations are fairly straightforward and will be skipped.

Theorem 1. If $n\alpha \to \lambda > 0$, as $n \to \infty$, then the p.g.f. of N_{I} converges to

$$\phi_{\mathrm{I}}(t) = [1 + p(1-\beta)^{k-1}\beta(t-1)]\exp\{\lambda(1-\beta)^{k-1}\beta(t-1)\}.$$

Thus the limiting distribution of N_{I} is the convolution of a Bernoulli with parameter $p(1-\beta)^{k-1}\beta$ and a Poisson with parameter $\lambda(1-\beta)^{k-1}\beta$.

Theorem 2. If $n\alpha \to \lambda > 0$, as $n \to \infty$, then the p.g.f. of N_{II} converges to

$$\phi_{\mathrm{II}}(t) = [1 + p(1-\beta)^{k-1}(t-1)]\exp\{\lambda(1-\beta)^{k-1}(t-1)\}.$$

Thus the limiting distribution of N_{II} is the convolution of a Bernoulli with parameter $p(1-\beta)^{k-1}$ and a Poisson with parameter $\lambda(1-\beta)^{k-1}$.

Theorem 3. If $n\alpha \to \lambda > 0$, as $n \to \infty$, then the p.g.f. of N_{III} converges to

$$\phi_{\mathrm{III}}(t) = \left\{1 - p(1-\beta)^{k-1} + \frac{p(1-\beta)^{k-1}(1-(1-\beta)^k)t}{1 - t(1-\beta)^k}\right\}$$

$$\times \exp\left\{\lambda(1-\beta)^{k-1}\left\{\frac{(1-(1-\beta)^k)t}{1-t(1-\beta)^k} - 1\right\}\right\}.$$

Thus the limiting distribution of N_{III} is the convolution of the mixture of a Bernoulli distribution with parameter $p(1-\beta)^{k-1}$ and a geometric distribution with parameter

$(1-\beta)^k$, and a compound Poisson distribution with parameter $\lambda(1-\beta)^{k-1}$ and geometric compounding distribution whose p.g.f. is $(1-(1-\beta)^k)t/[1-t(1-\beta)^k]$.

Theorem 4. If $n\alpha\to\lambda>0$, as $n\to\infty$, then the p.g.f. of N_{IV} converges to

$$\phi_{\mathrm{IV}}(t)=\left\{1-p(1-\beta)^{k-1}+\frac{p(1-\beta)^{k-1}\beta t}{1-t(1-\beta)}\right\}\exp\left\{\lambda(1-\beta)^{k-1}\left\{\frac{\beta t}{1-t(1-\beta)}-1\right\}\right\}.$$

Thus the limiting distribution of N_{IV} is the convolution of the mixture of a Bernoulli distribution with parameter $p(1-\beta)^{k-1}$ and a geometric distribution with parameter $(1-\beta)$, and a compound Poisson distribution with parameter $\lambda(1-\beta)^{k-1}$ and geometric compounding distribution whose p.g.f. is $\beta t/[1-(1-\beta)t]$.

We shall record the next results as Corollary 1 without proof. We call it a corollary because the completely stationary model (2) is a special case of the Markov model (1). But it is easy to see that it can not be obtained by substituting the parameters p and π in the four theorems. In the completely stationary case the three parameters p, α and β interweave, and the values of p and β change as $n\alpha$ tends to λ, while in the other Markov model all the three parameters act independently. Lemma 1 is still applicable to Corollary 1.

Corollary 1. Let $\{X_i\}$ be a completely stationary Markov Bernoulli sequence with transition probabilities defined by (2). If $np\to\lambda>0$, as $n\to\infty$, then the limiting distribution of:

1. N_{I} is a Poisson with parameter $\lambda(1-\pi)^2\pi^{k-1}$;
2. N_{II} is a Poisson with parameter $\lambda(1-\pi)\pi^{k-1}$;
3. N_{III} is a compound Poisson with parameter $\lambda(1-\pi)\pi^{k-1}$ and geometric compounding distribution $h(x)=(1-\pi^k)\pi^{(x-1)k}$, $x=1,2,\cdots$;
4. N_{IV} is a compound Poisson with parameter $\lambda(1-\pi)\pi^{k-1}$ and geometric compounding distribution $h(x)=(1-\pi)\pi^{x-1}$, $x=1,2,\cdots$.

It should be remarked here that for the asymptotically i.i.d. Markov Bernoulli model and the i.i.d. model, under the limit condition '$np\to\lambda>0$ as $n\to\infty$', the *limiting distributions of all the four variables N for $k>1$ are degenerate at* 0. The proper condition for meaningful limits for these two models is '$np^k\to\lambda>0$ as $n\to\infty$'. By following the proofs of the four theorems it can be shown that under this condition *the limiting distributions of all the four variables N are identical — Poisson with parameter λ*. (Lemma 1 is no longer applicable in this case. A different one is needed.) This result can be used to deduce many recent theorems in the literature such as Papastavridis (1987), Chao and Fu (1989), Godbole (1990), Fu (1993).

4. Discussion

In closing, we would like to add two points. One is an application to the runs test in statistics. The other is possible extensions.

(a) To test the hypothesis whether a sequence of n binary numbers is random, one popular method is to use the total number of runs R. (See Mood (1940).) The statistic R is related to N_{II} with $k=1$ as follows: if R is even, then $R=2N_{II}$; if R is odd, then $R=2N_{II}-1$, for $X_1=1$, and $R=2N_{II}+1$, for $X_1=0$. Thus if n_1, the number of 1's in the sequence of n binary numbers, is small relative to n, the asymptotic distribution H of R is

$$H(x)=\begin{cases} e^{-\lambda}\lambda^k/k! & \text{for } x=2k,\\ pe^{-\lambda}\lambda^{k+1}/(k+1)!+(1-p)e^{-\lambda}\lambda^{k-1}/(k-1)! & \text{for } x=2k+1,\end{cases}$$

where $\lambda=n_1p$ and $p=n_1/n$. (If $n-n_1$ is small relative to n, a similar asymptotic distribution of R can be derived accordingly.)

(b) Historically, parallel to the problem of 'runs' was the problem of 'patterns'. Since the definition of runs has expanded from Types I and II to cover Type III by Feller (1950) and then Type IV by Ling (1988), patterns should now be considered as special cases of runs. (See Wang and Liang (1993) for more detail.) Let us consider an example here. A combination padlock with a circular dial can be unlocked only if the dial is turned, from the center, say, to the left k_1 times and then to the right k_2 times. What is the probability of unlocking it if the dial is turned left and right in a random manner? This and other examples prompted Huang and Tsai (1989) to construct a 'binomial distribution of order (k_1, k_2)' which extended an earlier work by Hirano (1984). Their binomial distribution is the distribution of a mixture of Type I and Type II runs under the i.i.d. structure.

As anyone who has ever tried a combination padlock knows, a padlock can be unlocked only if it is turned *exactly* k_1 times to the left and then *exactly* k_2 times to the right. But if you want to reach a telephone number such as 848–3233 and mistakenly dial two extra digits such as 848–323333, it still works. Therefore we have two kinds of mixtures of runs introduced in Section 1. The first kind is 'a run of k_1 F of Type I followed by another run of k_2 S of Type I', the second kind is 'a run of k_1 F of Type I followed by another run of k_2 S of Type II'. We shall denote the number of occurrences of the first kind by M_I and that of the second kind by M_{II}, and call, under the i.i.d. structure, the distribution of M_I *the Type I binomial distribution of order* (k_1, k_2), and the distribution of M_{II} the *Type II binomial distribution of order* (k_1, k_2). Huang and Tsai's binomial distribution is of Type II.

To find the limits of both types of binomial distributions we define

$$W_i^{(I)}=I(V_i=k_1,\ U_i=k_2), \tag{11}$$

and

$$W_i^{(II)}=I(V_i=k_1,\ U_i\geqq k_2). \tag{12}$$

Using these two sequences of indicator variables, we define $\mathbb{S}_n^{(j)}$ by (5). Thus to derive the limiting distributions of M_I and M_{II} it is sufficient to derive those of $\mathbb{S}_n^{(I)}$ and $\mathbb{S}_n^{(II)}$. Under the i.i.d. structure, it can be shown that the limiting distribution of M_I is Poisson (λ), if $\lim np^{k_2+1}(1-p)^{k_1}\to\lambda$, $(n\to\infty)$, and that of M_{II} is also Poisson (λ) if

$np^{k_2}(1-p)^{k_1} \to \lambda$, $(n \to \infty)$. (See also Huang and Tsai (1991), Corollary 1.) Under the other two Markovian structures, it can be shown that their limits are also Poisson.

Evidently the limiting distributions of the number of occurrences of other kinds of runs mixtures can be derived accordingly.

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